

Dr. Chilperic Armel Foko Kuate

Computational Systems Biologist | Applied Mathematician | Mechanistic Modeling

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★ RESEARCH AREAS & METHODS

- Systems biology and mechanistic modelling
- Hepatic metabolism and lipid dynamics
- Plant physiology and photosynthetic adaptation
- Quantitative immunology and T-cell dynamics
- ODE, DAE, and stochastic models
- Parameter estimation, calibration, and uncertainty analysis
- Constraint-based and differentiable metabolic modeling

</> PROGRAMMING & TOOLS

- </> Languages
Python, Julia, R, MATLAB
- ⚙️ Optimisation
CasADi, IPOPT, CMA-ES, JuMP, Pyomo
- 📊 Scientific Computing
SciPy, NumPy, CVODE, SUNDIALS, lmfit
- 📈 Data & Visualisation
pandas, matplotlib, ggplot2, Plotly, Bokeh
- 🔍 Sensitivity Analysis
SALib — Sobol, Morris
- 🏠 Applied ML
regression, classification; scikit-learn
- >_ Workflow
git, Jupyter, 🐧 Linux, HPC
- 🌐 Web Tools
HTML, CSS, JavaScript, browser-based interfaces, interactive simulation tools

LANGUAGES

- French Native
- English Fluent
- German B1+

🏆 AWARDS & FELLOWSHIPS

- ★ Bayer B4U Mentoring Programme, mentee (2026)
- ★ Marie Curie ITN Fellowship, PoLiMeR (2019–2023)
- ★ Post-AIMS Research Bursary (2017–2019)
- ★ F.K.A. Allotey Meritorious Award (2016)

👤 PROFILE

Computational biologist and applied mathematician building mechanistic models of biological systems, from hepatic metabolism and immune-cell dynamics to photosynthetic adaptation across the C3 to C4 continuum. My work combines dynamical systems theory, stochastic modeling, and biological interpretation, with a focus on what mathematics can explain about living systems, not just fit. I also develop scientific and educational web tools that make simulation, analysis, and technical training more accessible.

🧰 RESEARCH EXPERIENCE

Independent Research, Scientific Software, and Further Training Nov 2025 – Present *Düsseldorf, Germany*

- Preparing two manuscripts from postdoctoral research on the C3 to C4 continuum: a first-principles thermo-hydraulic framework for photosynthetic adaptation and a structure-function framework for leaf photosynthesis.
- Developed browser-based tools for scientific modeling and language learning, including Foko Lab and Deutsch-WiPA 2026.
- Pursuing structured upskilling in AI engineering and professional German to strengthen research-software development and scientific work in German-speaking environments.

Postdoctoral Researcher Nov 2022 – Oct 2025 *CEPLAS / Institute of Computational Cell Biology, HHU Düsseldorf*

- Developed a first-principles thermo-hydraulic-biochemical modeling framework for photosynthetic adaptation across the C3 to C4 continuum in *Flaveria*, linking mesophyll and bundle-sheath physiology to soil and environment.
- Coupled C3 to C4 FvCB biochemistry, heat balance, hydraulic regulation, and environmental forcing to explain how anatomy, transport, and physiology jointly shape photosynthetic performance.
- Applied Sobol sensitivity analysis and CMA-ES-based evolutionary simulations across contrasting climatic regimes to identify influential traits, quantify trade-offs, and show that different climates favor different adaptive routes rather than a single universal optimum.
- Presented this work at the CEPLAS Early Career Researcher Symposium, Düsseldorf, 2025.
- Supervised 2 MSc and 2 BSc projects in computational plant biology.

Marie Curie PhD Fellow Aug 2019 – Aug 2023 *Institute of Quantitative and Theoretical Biology, HHU Düsseldorf*

- Developed dynamic and stochastic models of hepatic fatty-acid metabolism to clarify regulatory mechanisms underlying metabolite dynamics and disease-relevant metabolic behavior.
- Built a coarse-grained model of fatty-acid metabolism and derived conditions for bi-stability, providing a mechanistic explanation for metabolic switching between fed and fasted states.
- Developed a semi-mechanistic model of fatty-acid de novo synthesis, fitted it to time-course data, and showed when production rates can be approximated by Michaelis–Menten laws, improving kinetic interpretability.
- Performed parameter estimation, sensitivity analysis, and uncertainty analysis for nonlinear biological systems to strengthen mechanistic interpretation of hepatic fatty-acid metabolism.
- Developed the enzyme-kinetic component of a differentiable constraint-based metabolic modeling framework, formalizing how kinetic constraints integrate into systems-level metabolic analysis; published in *Metabolic Engineering* (2022).

Research Fellow Nov 2017 – Mar 2019 *AIMS Cameroon & AIMS Rwanda*

- Modeled T-cell proliferation dynamics using stochastic and deterministic frameworks, with emphasis on branching processes, individual-based simulation, and moment-based analysis.
- Developed a stochastic master-equation framework for T-cell proliferation, derived mean and variance expressions for cell counts, and set up a maximum-likelihood fitting framework for CFSE-type data.

- ★ Best Student, AIMS Ghana (2016)
- ★ AIMS Ghana Hackathon Winner (2016)

🎓 EDUCATION

PhD, Quantitative & Theoretical Biology
HHU Düsseldorf 2019–2023

Master 2 Mathematics & Applications
Univ. Yaoundé I 2017–2019

MSc Mathematical Sciences
AIMS Ghana 2015–2016

BSc Mathematics
Univ. Douala 2009–2012

🔧 TEACHING & SUPERVISION

Supervised 2 MSc and 2 BSc projects in computational biology (HHU, 2019–2025).

Delivered 200+ teaching hours — mathematical modelling, ODEs, stochastic processes, Python (AIMS Rwanda, AIMS Cameroon, HHU).

💻 SOFTWARE & WEB TOOLS

Foko Lab
Browser-based environment for ODE, optimisation, steady-state, and stochastic modeling workflows.

🔗 [Open web tool](#)

Deutsch-WiPA 2026
Interactive B1 to B2 German learning tool with conjugation practice, feedback loops, and error tracking.

🔗 [Open web tool](#)

👥 REFERENCES Prof. Dr.

Martin J. Lercher
HHU Düsseldorf
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Prof. Dr. Oliver Ebenhöh
HHU Düsseldorf
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Prof. Dr. Wilfred Ndifon
AIMS Global Network
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Prof. Dr. Barbara M. Bakker
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🎓 SELECTED TRAINING & CONFERENCES 🏆 Bayer B4U Mentoring Programme (Mentee) Feb 2026

Inaugural edition in Germany; one-to-one pairing with Bayer scientists.

🌱 **PoLiMeR Marie Curie ITN Doctoral Training** 2020–2022
Training across metabolism, regulation, enzyme catalysis, and scientific communication.

💻 **Advanced AI and Data Science Workshop** Dec 2021
HeiCAD, HHU Düsseldorf — neural networks, optimisation, and applied data science tools.

🔗 **AI Engineering Learning Track** 2025–Present
Ongoing self-directed training in AI engineering, applied machine learning, and technical implementation workflows.

🇩🇪 **Professional German Development** Ongoing
Continued German progression beyond B1+ toward stronger professional fluency in academic and technical settings.

🌱 **Clinic on Meaningful Modelling of Epidemiological Data** Jul 2018
AIMS South Africa and SACEMA — epidemiological modeling and data-driven inference.

Talks

- Oct 2025 CEPLAS Early Career Researcher Symposium, Düsseldorf, *talk*
- Mar 2022 PoLiMeR Symposium, Düsseldorf, *talk & poster*
- Jul 2019 Theoretical Biology Seminar, Univ. Rwanda
- Aug 2018 Mathematics Against Malaria, Mit-tweida

Posters

- Jul 2023 Economic Principles in Cell Biology Summer School
- May 2022 Systems Biology of Mammalian Cells, Heidelberg
- Jul 2020 International Systems Biology Group, Innsbruck
- Jul 2018 MMED Epidemiological Modelling, Cape Town

📄 PUBLICATIONS Published

- **Foko Kuate C.A.**, Ebenhöh O., Bakker B.M., Raguin A. *Kinetic Rate Laws of the Enzymes Involved in Animal Fatty Acid Synthesis. Bioscience Reports*, 2023.
- Wilken S.E., Besançon M., Kratochvíl M., **Foko Kuate C.A.**, Trefois C., Gu W., Ebenhöh O. *Interrogating the Effect of Enzyme Kinetics on Metabolism Using Differentiable Constraint-Based Models. Metabolic Engineering*, 2022.
- Dourado H., **Foko Kuate C.A.**, Liebermeister W., Lercher M.J. *Growth Mechanics and the Emergence of Metabolic Oscillations in Growing Cells. bioRxiv*, 2025.

+2 manuscripts in preparation on plant adaptation across the C3–C4 continuum.